2007 Toyota Yaris T-bones School Bus Intersection Crash

Student Copy

A small school bus came to an intersection with a stop sign, stopped, then proceeded forward. A 2007 Toyota Yaris on the perpendicular crossing street, with no traffic control device other than the 25 mph speed limit, struck the front passenger wheel of the school bus in a classic 90 degree intersection crash. The primary issue in the case is who had the right of way.



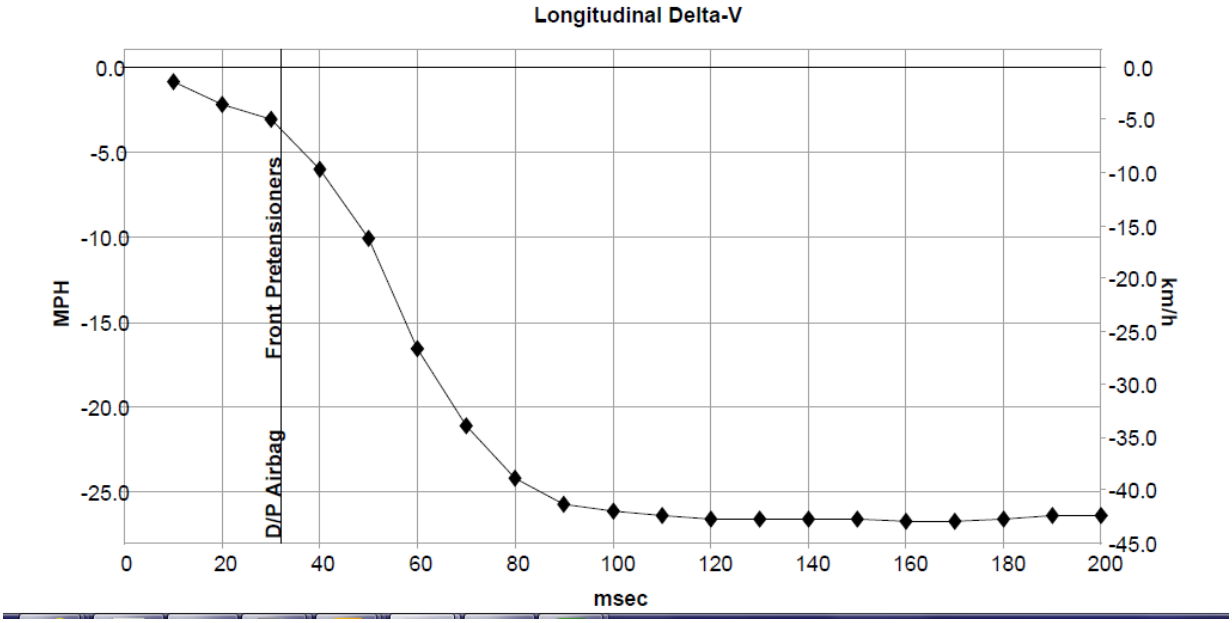
The Yaris is believed to have swerved right just before impact, and traveled 17 feet post impact to its point of rest, with one front wheel locked and another partially obstructed, for 40% effective braking yielding on a 0.7G drag factor road (range 0.65 to 0.75) yielding an equivalent drag factor of 0.28 (range 0.26 to 0.30) The Yaris departure angle was 31 degrees right of its approach angle. Your reconstruction is complicated by the fact the bus struck a curb and a telephone pole as it came to rest.

Review the attached CDR report and answer the following questions:

1. **Is the most recent event in this CDR report from the crash pictured above? What data elements are you relying upon to make this judgment?**

Longitudinal Crash Pulse (Most Recent Event, TRG 2 - table 1 of 2)

Recording Status, Time Series Data	Complete
Max Longitudinal Delta-V (MPH [km/h])	-26.7 [-43.0]



2. **Does the prior event recorded have anything to do with the crash under investigation?**

System Status at Time of Retrieval

ECU Part Number	89170-52780
ECU Generation	04EDR
Recording Status, All Pages	Complete
Freeze Signal	ON
Freeze Signal Factor	Front Airbag Deployment Front Pretensioner Deployment
Diagnostic Trouble Codes Exist	No
Time from Previous Pre Crash TRG (msec)	16381 or greater
Latest Pre-Crash Page	1
Contains Unlinked Pre-Crash Data	No

3. **The reported speed drops from 38.5 mph to 0 between -0.4 seconds to impact and the Trigger. How can this happen?**

Pre-Crash Data, -5 to 0 seconds (Most Recent Event, TRG 2)

Time (sec)	-4.4	-3.4	-2.4	-1.4	-0.4	0 (TRG)
Vehicle Speed (MPH [km/h])	57.2 [92]	55.9 [90]	52.2 [84]	52.2 [84]	38.5 [62]	0 [0]
Brake Switch	OFF	ON	OFF	ON	ON	ON
Accelerator Rate (V)	0.94	0.78	0.78	0.78	0.86	0.78
Engine RPM (RPM)	2,400	2,000	1,600	1,600	800	400

4. **Using only the speed versus time data, what is the range of possible speeds at impact?**

	MIN	MAX
Last **Good** Data Point		
Additional Braking Prior to Impact (0.4+0.5) sec x 15 mph/sec		+0
Slip Adjustment (brake under reporting 5%)	+	+
Speedometer Error +/-4%	-	+
	____	____

Range of possible speeds at impact

5. **Using only the Delta V data, what is the range of possible speeds at impact?**

   Hint: Speed at Impact $= V_3 Cos(\beta) - \Delta V_1$

   Apply +/-10% tolerance to Delta V, get **range ____mph to ____mph**

6. **Considering both the speed data and the Delta V in the EDR, what is the range of possible speeds at impact of the Yaris? (Combine the ranges)**

Range From Speed Data: _____
Range from Delta V _____
Combined Range _____
(In the real world you would also combine the range from the scene evidence, if you could get one)

7. **According to the EDR, what was the fastest the Yaris was traveling in the 5 seconds before the crash?**

_____mph at -____ seconds to impact, plus or minus 4% speed data tolerance, yields a range of _____ **mph**

8. **If the Yaris driver had been going the speed limit, would the crash have occurred?**
   Hint: No detailed calculations are required, make an approximate judgment yes or no assuming the Yaris is going an average of at least 50mph. FYI, human factors studies tell us that in a 25 mph speed zones the average driver will pull out to cross an intersection if they perceive they have a gap in traffic equal to 5.6 seconds of travel at the speed limit. Using a standard acceleration factor for the bus, the bus driver is believed to have started accelerating from the stop sign at 4.5 seconds before impact. The bus needed about another 20 feet to clear the intersection.

IMPORTANT NOTICE: Robert Bosch LLC and the manufacturers whose vehicles are accessible using the CDR System urge end users to use the latest production release of the Crash Data Retrieval system software when viewing, printing or exporting any retrieved data from within the CDR program. Using the latest version of the CDR software is the best way to ensure that retrieved data has been translated using the most current information provided by the manufacturers of the vehicles supported by this product.

CDR File Information

User Entered VIN	JTDBT923171******
User	Wendell R Cover
Case Number	11.2529
EDR Data Imaging Date	09/12/2011
Crash Date	05/18/2011
Filename	07 YARIS BUS CRASH CASE SETTLED 04 EDR.CDRX
Saved on	Monday, September 12 2011 at 19:49:58
Collected with CDR version	Crash Data Retrieval Tool 4.1.1
Reported with CDR version	Crash Data Retrieval Tool 12.0
EDR Device Type	Airbag Control Module
Event(s) recovered	Front/Rear (2)

Comments

2007 toyota yaris
acm benchtop

Data Limitations

CDR Record Information:

- Due to limitations of the data recorded by the airbag ECU, such as the resolution, data range, sampling interval, time period of the recording, and the items recorded, the information provided by this data may not be sufficient to capture the entire crash.
- Pre-Crash data is recorded in discrete intervals. Due to different refresh rates within the vehicle's electronics, the data recorded may not be synchronous to each other.
- Airbag ECU data should be used in conjunction with other physical evidence obtained from the vehicle and the surrounding circumstances.
- If the airbags did not deploy or the pretensioners did not operate during an event that meets a specified recording threshold, it is called a Non-Deployment Event. Data from a Non-Deployment Event can be overwritten by a succeeding event that meets the specified recording threshold. If the airbag(s) deploy or the pretensioners are operated, it is called a Deployment Event. Deployment Event data cannot be overwritten or deleted by the airbag ECU following that event.
- If power supply to the airbag ECU is lost during an event, all or part of the data may not be recorded.
- "Diagnostic Trouble Codes" are information about faults when a recording trigger is established. Various diagnostic trouble codes could be set and recorded due to component or system damage during an accident.
- The airbag ECU records only diagnostic information related to the airbag system. It does not record diagnostic information related to other vehicle systems.
- The TaSCAN, Global TechStream, or Intelligent Tester II devices (or any other Toyota genuine diagnostic tool) can be used to obtain detailed information on the diagnostic trouble codes from the airbag system, as well as diagnostic information from other systems. However, in some cases, the diagnostic trouble codes of the airbag system recorded by the airbag ECU when the event occurred may not match the diagnostic trouble codes read out when the diagnostic tool is used.

General Information:

- The data recording specifications of Toyota's airbag ECUs are divided into the following seven categories. The specifications for 12EDR or later are designed to be compatible with NHTSA's 49CFR Part 563 rule.
  - 00EDR / 02EDR / 04EDR / 06EDR / 10EDR / 12EDR / 13EDR
- The airbag ECU records data for all or some of the following accident types: frontal crash, rear crash , side crash, and rollover events. Depending on the installed airbag ECU, data for side crash and/or rollover events may not be recorded.
- This airbag ECU records post-crash data, and depending on the airbag ECU, may record pre-crash data.
  - If a single event occurs independently, the data for that event is recorded on a one-to-one basis.
  - If multiple events occur successively (within a period of approximately 500ms), the establishment of the recording trigger for the first event is defined as the "pre-crash recording trigger". Pre-crash data for the first event and post-crash data for each successive event is then recorded.
  - In some situations, pre-crash data may be re-recorded if an event continues for more than 0.5 seconds.
- The airbag ECU has two recording pages (memory maps) to store pre-crash data. Additionally, to store post-crash data, the airbag ECU has two recording pages for each accident type: two pages for frontal and rear crash, two pages for a side crash, and two pages for rollover event.
- The data recorded by the airbag ECU includes correlating information between each previously occurring event (i.e., information that clarifies the collision event sequence. This correlation information consists of the following items.
  - Time from Previous Pre-Crash TRG
  - Linked Pre-Crash Page
  - Time from Pre-Crash TRG

- TRG Count
- Previous Crash Type

- The point in time at which the recording trigger is established is regarded as time zero for the recorded data.
- The recording trigger judgment threshold value differs depending on the collision type (i.e., frontal crash, rear crash, side crash, or rollover event).
- Time series data for side crash may have 24 or 25 sampling points.
- Some of the data recorded by the airbag ECU is transmitted to the airbag ECU from various vehicle control modules by the vehicle's Controller Area Network (CAN).
- In some cases, the airbag ECU part number printed on the ECU label may not match the airbag ECU part number that the CDR tool reports. The part number retrieved by the CDR tool should be considered as the official ECU part number.

Data Element Sign Convention:
The following table provides an explanation of the sign notation for data elements that may be included in this CDR report.

Data Element Name	Positive Sign Notation Indicates
Max. Longitudinal Delta-V	Forward
Longitudinal Delta-V	Forward
Max. Lateral Delta-V , B-Pillar Sensor	Outside to Inside
Max. Lateral Delta-V , C-Pillar Sensor	Outside to Inside
Max. Lateral Delta-V , Slide Door Sensor	Outside to Inside
Lateral Delta-V , B-Pillar Sensor	Outside to Inside
Lateral Delta-V , C-Pillar Sensor	Outside to Inside
Lateral Delta-V , Slide Door Sensor	Outside to Inside
Lateral Delta-V , Airbag ECU Sensor	Outside to Inside
Roll Angle Peak	Clockwise Rotation
Roll Angle	Clockwise Rotation
Lateral Acceleration , Airbag ECU Sensor *	Right to Left

* For sensing a rollover

Data Definitions:
1)

- The "ON" setting for the "Freeze Signal" indicates a state in which the non-volatile memory can not be overwritten or deleted by the airbag ECU. After "Freeze Signal" has been turned ON, subsequent events will not be recorded.
- "Recording Status" indicates a state in which all recorded event data has been written into the non-volatile memory, or a state in which this process was interrupted and not fully written into the non-volatile memory. If "Recording Status" is "Incomplete", recorded event data may not be valid.
- "Time to Deployment Command" indicates the time between recording trigger establishment and the determination of airbag deployment. This value may differ from the actual time it takes for the airbag to fully deploy.
- Even if an airbag/pretensioner did not deploy due to the "front passenger airbag disable switch and/or "RSCA Disable Switch" in the ON position or other disabling criteria are met, the "Time to deployment command" data element for that airbag/pretensioner may still be recorded.
- "Engine RPM" indicates the number of engine revolutions, not the number of motor revolutions. The recorded value has an upper limit of 5,200 rpm. Resolution is 400 rpm and the value is rounded down and recorded. For example, if the actual engine speed is 799 rpm, the recorded value will be 400 rpm.
- The upper limit for the recorded "Vehicle Speed" value is 122 km/h (75.8mph). Resolution is 2km/h (1.2mph) and the value is rounded down and recorded. The accuracy of the "Vehicle Speed" value can be affected by various factors. These include, but not limited, to the following.
  - Significant changes in the tire's rolling radius
  - Wheel lock and wheel slip
- "Accelerator Rate" value is recorded as a voltage. The voltage increases as the driver depresses the accelerator.
- The "Drive" setting for the "Shift Position" value indicates the shift position state is other than "R,"(Reverse), "N" (Neutral), or "P" (Park).
- Depending on the type of occupant sensor installed in the vehicle, one of the following three recording formats for "Occupancy Status, Passenger" will be utilized.
  - Occupied / Not Occupied
  - Adult / Child / Not Occupied
  - AM50 / AF05 / Child / Not Occupied
- "Ignition Cycle Since DTC was Set" records the cumulative amount of times that the ignition is switched ON after an airbag system diagnostics code was set for the first time.
- "Air Bag Warning Lamp ON Time Since DTC was Set" records the total time that the ignition has been switched ON after the warning lamp was illuminate, due to an airbag system fault, for the first time. The resolution is 15 minutes, and the value is rounded down and recorded.
- "Longitudinal Delta-V" indicates the change in forward speed after establishment of the recording trigger. This does not refer to vehicle speed, and it does not include the change in speed during the period from the start of the actual collision to establishment of the recording trigger.
- "Roll Angle peak" may not always match the peak value within the "Roll Angle" sampling points due to differences in data calculation method.
- For "Lateral Delta-V", the sensor location (B-pillar, front door, C-pillar, and slide door) shows the outline of a typical sensor position. Sensory location can be confirmed using the repair manual.
- "Time from Previous Pre-Crash TRG" indicates the time between the establishment of an event's pre-crash recording trigger to the

establishment of a more recent event's pre-crash recording trigger. The upper limit for the recorded value is 16,381 milliseconds. In the event of establishment of the first pre-crash recording trigger after the ignition is switched ON, the upper limit value(max value) is recorded.

- "TRG Count" indicates a calculated value of the number of times recording triggers have been established for all crash types. The sequence in which each event occurred can be verified from the "TRG Count". The smaller the "TRG Count" value, the older the data. The upper limit for the recorded value is 65,533 times. When more than one event reaches the upper limit, the actual "TRG Count" may be greater than what is displayed for that event.
- "Linked Pre-Crash Page" is used to link 'paged" pre-crash data with 'paged" post-crash data. When old pre-crash data is overwritten by new pre-crash data, the "Linked Pre-Crash Page" value may record a page number that is not actually linked.
- Resolution of the "Time from Pre-Crash to TRG" is 100 [ms], and the value is rounded down and recorded.

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System Status at Time of Retrieval

ECU Part Number	89170-52780
ECU Generation	04EDR
Recording Status, All Pages	Complete
Freeze Signal	ON
Freeze Signal Factor	Front Airbag Deployment Front Pretensioner Deployment
Diagnostic Trouble Codes Exist	No
Time from Previous Pre Crash TRG (msec)	16381 or greater
Latest Pre-Crash Page	1
Contains Unlinked Pre-Crash Data	No

Event Record Summary at Retrieval

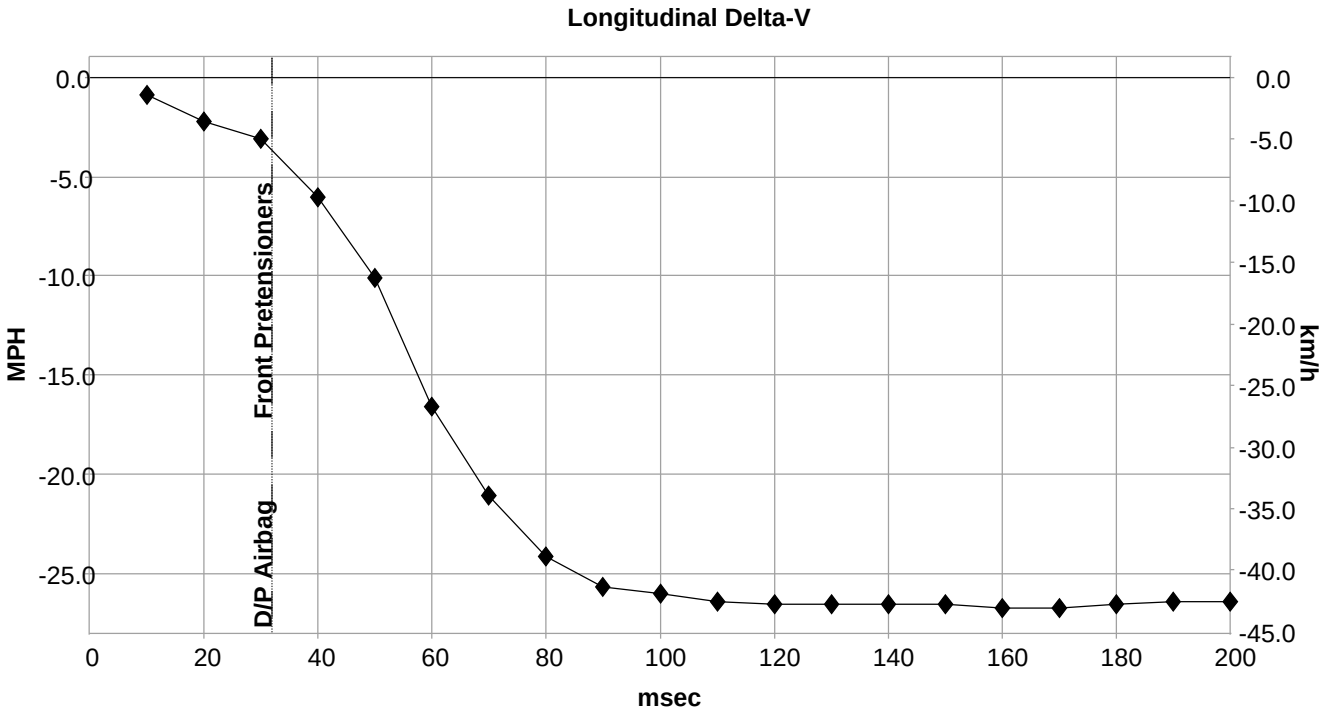
Events Recorded	TRG Count	Crash Type	Time (msec)	Pre-Crash & DTC Data Recording Status	Event & Crash Pulse Data Recording Status
Most Recent Event	2	Front/Rear Crash	0	Complete (Page 1)	Complete (Front/Rear Page 1)
1st Prior Event	1	Front/Rear Crash	-16381 or greater	Complete (Page 0)	Complete (Front/Rear Page 0)

System Status at Event (Most Recent Event, TRG 2)

Recording Status, Front/Rear Crash Info.	Complete
Crash Type	Front/Rear Crash
TRG Count (times)	2
Previous Crash Type	No Event
Time from Pre-Crash TRG (msec)	0
Linked Pre-Crash Page	1
Time to Deployment Command, Front Airbag, Driver (msec)	32
Time to Deployment Command, Front Airbag, Passenger (msec)	32
Event Severity Status, Driver	Level 3
Event Severity Status, Passenger	N/A
Time to Deployment Command, Pretensioner (msec)	32

Longitudinal Crash Pulse (Most Recent Event, TRG 2 - table 1 of 2)

Recording Status, Time Series Data	Complete
Max Longitudinal Delta-V (MPH [km/h])	-26.7 [-43.0]



Longitudinal Crash Pulse (Most Recent Event, TRG 2 - table 2 of 2)

Time (msec)	Longitudinal Delta-V (MPH [km/h])
10	-0.9 [-1.4]
20	-2.2 [-3.6]
30	-3.1 [-5.0]
40	-6.0 [-9.7]
50	-10.1 [-16.3]
60	-16.6 [-26.8]
70	-21.1 [-33.9]
80	-24.2 [-38.9]
90	-25.7 [-41.4]
100	-26.1 [-41.9]
110	-26.4 [-42.5]
120	-26.6 [-42.8]
130	-26.6 [-42.8]
140	-26.6 [-42.8]
150	-26.6 [-42.8]
160	-26.7 [-43.0]
170	-26.7 [-43.0]
180	-26.6 [-42.8]
190	-26.4 [-42.5]
200	-26.4 [-42.5]

DTCs Present at Time of Event (Most Recent Event, TRG 2)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

Pre-Crash Data, 1 Sample (Most Recent Event, TRG 2)

Recording Status, Pre-Crash/Occupant	Complete
Time from Pre-Crash to TRG (msec)	400
Buckle Switch, Driver	Buckled
Buckle Switch, Passenger	Unbuckled
Occupancy Status, Passenger	Not Occupied
Seat Position, Driver	Rearward
Shift Position	Drive

Pre-Crash Data, -5 to 0 seconds (Most Recent Event, TRG 2)

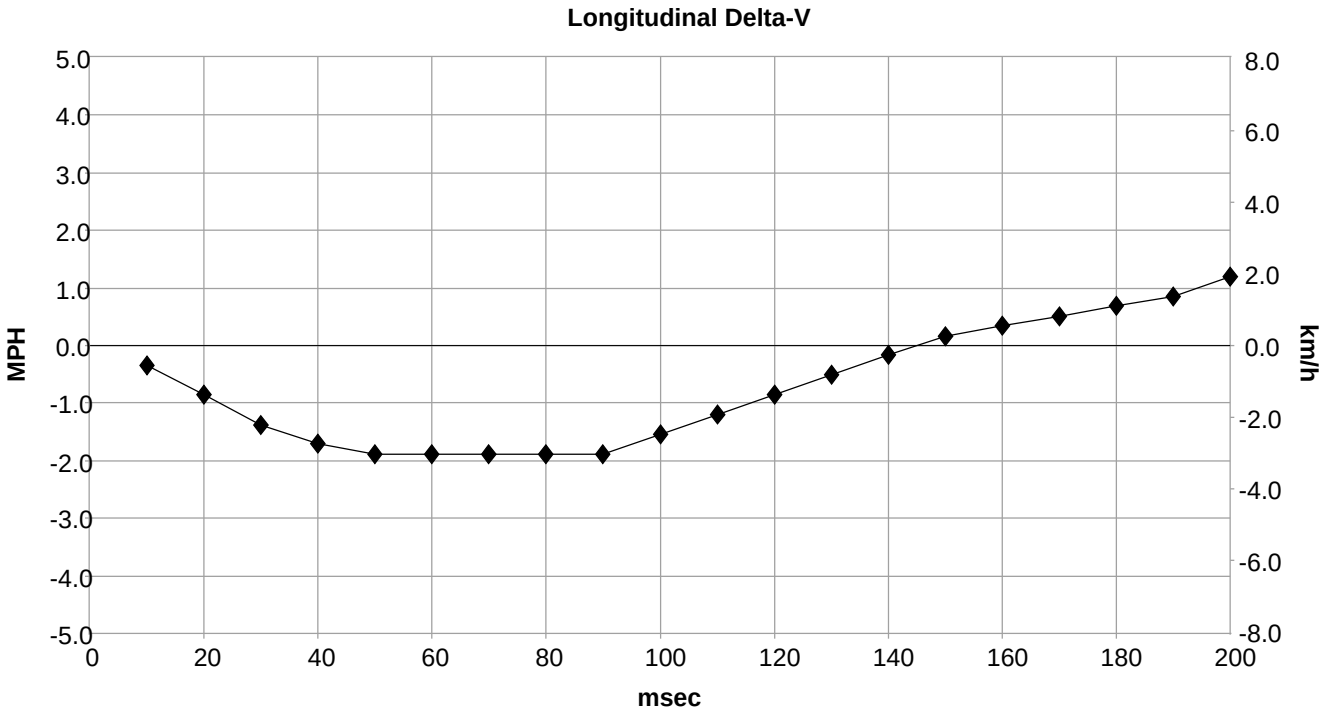
Time (sec)	-4.4	-3.4	-2.4	-1.4	-0.4	0 (TRG)
Vehicle Speed (MPH [km/h])	57.2 [92]	55.9 [90]	52.2 [84]	52.2 [84]	38.5 [62]	0 [0]
Brake Switch	OFF	ON	OFF	ON	ON	ON
Accelerator Rate (V)	0.94	0.78	0.78	0.78	0.86	0.78
Engine RPM (RPM)	2,400	2,000	1,600	1,600	800	400

System Status at Event (1st Prior Event, TRG 1)

Recording Status, Front/Rear Crash Info.	Complete
Crash Type	Front/Rear Crash
TRG Count (times)	1
Previous Crash Type	No Event
Time from Pre-Crash TRG (msec)	0
Linked Pre-Crash Page	0
Time to Deployment Command, Front Airbag, Driver (msec)	Not Commanded
Time to Deployment Command, Front Airbag, Passenger (msec)	Not Commanded
Event Severity Status, Driver	N/A
Event Severity Status, Passenger	N/A
Time to Deployment Command, Pretensioner (msec)	Not Commanded

Longitudinal Crash Pulse (1st Prior Event, TRG 1 - table 1 of 2)

Recording Status, Time Series Data	Complete
Max Longitudinal Delta-V (MPH [km/h])	-1.9 [-3.0]



Longitudinal Crash Pulse (1st Prior Event, TRG 1 - table 2 of 2)

Time (msec)	Longitudinal Delta-V (MPH [km/h])
10	-0.3 [-0.6]
20	-0.9 [-1.4]
30	-1.4 [-2.2]
40	-1.7 [-2.8]
50	-1.9 [-3.0]
60	-1.9 [-3.0]
70	-1.9 [-3.0]
80	-1.9 [-3.0]
90	-1.9 [-3.0]
100	-1.5 [-2.5]
110	-1.2 [-1.9]
120	-0.9 [-1.4]
130	-0.5 [-0.8]
140	-0.2 [-0.3]
150	0.2 [0.3]
160	0.3 [0.6]
170	0.5 [0.8]
180	0.7 [1.1]
190	0.9 [1.4]
200	1.2 [1.9]

DTCs Present at Time of Event (1st Prior Event, TRG 1)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Airbag Warning Lamp ON Time Since DTC was Set (min)	0
Diagnostic Trouble Codes	None

Pre-Crash Data, 1 Sample (1st Prior Event, TRG 1)

Recording Status, Pre-Crash/Occupant	Complete
Time from Pre-Crash to TRG (msec)	900
Buckle Switch, Driver	Unbuckled
Buckle Switch, Passenger	Unbuckled
Occupancy Status, Passenger	Not Occupied
Seat Position, Driver	Rearward
Shift Position	Drive

Pre-Crash Data, -5 to 0 seconds (1st Prior Event, TRG 1)

Time (sec)	-4.9	-3.9	-2.9	-1.9	-0.9	0 (TRG)
Vehicle Speed (MPH [km/h])	1.2 [2]	1.2 [2]	2.5 [4]	3.7 [6]	5 [8]	5 [8]
Brake Switch	OFF	OFF	OFF	OFF	OFF	OFF
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	400	400	800	800	800	800

Hexadecimal Data

Data that the vehicle manufacturer has specified for data retrieval is shown in the hexadecimal data section of the CDR report. The hexadecimal data section of the CDR report may contain data that is not translated by the CDR program. The control module contains additional data that is not retrievable by the CDR system.

PIDs	PID	Data
	00	BC 00 00 01
	01	00
	03	35 32 37 38 30 30 30 30 30 30 30 30 30 30 30
	04	FF FF FF FF
	05	02
	06	02
	20	80 00 C0 01
	21	00 11
	40	80 00 00 01
	41	3B 0B 00 D3 00 00 00 00
	60	00 00 00 01
	80	00 00 00 01
	A0	00 00 00 01
	C0	00 00 00 01
	E0	C0 10 00 00
	E1	0E
	E2	00 5B 1F 11 00
	EC	00

EEPROM	Address	Data
	0	00 00 00 00 42 00 78 4A 3D 48 32 40 3A 36 71 06
	10	00 00 00 00 42 00 78 4A 3D 48 32 40 3A 36 71 06
	20	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
	30	00 00 00 00 7F FD 00 00 A5 03 00 02 00 00 00 00
	40	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
	50	00 00 00 01 01 10 14 22 10 14 22 0C 14 11 08 14
	60	04 14 04 14 09 00 00 00 00 00 00 00 00 00 00 00
	70	00 00 00 00 00 00 00 00 00 00 10 01 01 01 14 12
	80	7D 16 44 A9 14 56 A8 14 B5 14 B8 18 04 00 00 00
	90	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
	A0	00 00 00 00 02 03 03 02 01 00 00 00 00 FE FE FE
	B0	FE FE FE FF FF FF FF FE 00 6E 00 00 00 01 E0 00
	C0	FE FE 00 00 05 08 05 11 18 26 1A 12 09 02 02 01
	D0	00 00 00 01 00 FF FF 00 05 10 00 00 00 02 E4 00
	E0	20 20 30 00

Disclaimer of Liability

The users of the CDR product and reviewers of the CDR reports and exported data shall ensure that data and information supplied is applicable to the vehicle, vehicle's system(s) and the vehicle ECU. Robert Bosch LLC and all its directors, officers, employees and members shall not be liable for damages arising out of or related to incorrect, incomplete or misinterpreted software and/or data. Robert Bosch LLC expressly excludes all liability for incidental, consequential, special or punitive damages arising from or related to the CDR data, CDR software or use thereof.

EDR Analyst Guide

1. Are there any EDR's available from any of the vehicles involved that are supported by CDR? (Check help file)
2. Is there more than one EDR in any of the accident vehicles? Ford PCM or GM ROS? (Check help file)
3. Are there any EDR's not covered by CDR that I can get the manufacturer to read for me or read by another tool?

How MANY different events can be identified based on physical evidence in this crash? 1 2 3 3+ (Rollover/indeterminate)

How MANY events is this EDR capable of capturing? 1 2 3 4 5 6 ?? Types??

How MANY events has this EDR Captured? 1 2 3 4 5 6 ? Types: D ND DLE Front/Rear Side Rollover
What is the RECORDING THRESHOLD? Wakeup /2G? 5mph Don't Know? RECORDING PRIORITY? Most Recent? Largest?
For **EACH** event, determine if the event is from your crash and the data is usable

1. Is it a complete and valid recording? **Yes** (it says so) **No (says not) Doesn't Say – check DL for other criteria**
   Any evidence of power loss at impact? YES NO INDETERMINATE BATTTERY? FUSE BOX(s)? Shorts/Blown Fuses?
   If power loss, is this a 563 intent recorder with backup power supply for recording? YES NO NOT SURE
2. Key Cycles or On Time match? _____@event _____@imaging (for EACH event)
3. Delta V Large and/or consistent with visible crush or momentum analysis within 20%? YES NO INDETERMINATE N/A
4. Speed data consistent with reported facts, crush and postcrash travel? YES NO INDETERMINATE N/A

How much data do I have to work with?

Yes No Pre-crash speed and throttle _____ points at _____second intervals
Yes No Pre-crash braking _____ points at _____second intervals
Yes No Post crash data _____ points at _____second intervals
Yes No Any Delta V? Check off how much of that info is in the CDR report below
____ Cumulative ***Longitudinal***? (A single peak value listed in the report) at ____ ms after wakeup
____ Cumulative **Lateral**? (A single peak value listed in the report) at ____ms after wakeup
____ Longitudinal DV **Graph**? Duration ____ to ____ ms Zero = deploy or algorithm wakeup or other?

____ Lateral DV **Graph**? Duration ____ to ____ ms
____ If no Delta V, Acceleration data (Chrysler)? Duration ____ to ____ ms. Zero=deploy or wakeup?
Yes No Belt Buckled data entries in report? If present, Driver Buckled / Unbuckled? Pass. Buckled / Unbuckled?
Yes No Other things this EDR has that matter in this case?
_____ ABS on/off _____Traction Control on/off ____Stability Control on/off ____Speed Control on/off
_____ Steering Angle ___ Yaw Rate ___Roll Rate/Angle ____Lat Accel from Stab.Cont? ___Long. Accel
_____ Passenger Size Empty/Child/Adult? _____ Wheel RPM? _____ Tire Pressure? ____Other?

What is most ***important*** in this case?		Value As recorded In EDR	Adjustment required? See checklist	Adjusted Value or range?	
Yes No Speed before start of event	If you have it, enter it here	_____	Yes No	_____	+/-4%
Yes No Speed at Impact	If you have it, enter it here	_____	Yes No	_____	(see wksht)
Yes No Post impact exit velocity	If you have it, enter here:	_____	Yes No		
Yes No Delta V	If you have it, enter it here	____X____Y	Yes No	_____	+/-10%
Yes No _____	If you have it, enter it here	_____			

What else? Inspect the entire report – evidence of loss of control, swerve before impact?
Speed changes over time greater that allowed by laws of physics without an impact or special circumstances?
(May not be necessary if it does it matter to your main conclusions).

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Last updated May 11, 2013

EDR Analyst Guide

Are there any *SPECIAL CIRCUMSTANCES* in this case that could affect the EDR data accuracy?
(circle Yes/No/Don't Know)

Speed Data

Vehicle Operational Conditions

Yes No Don't Know 1. Wheels locked by braking without ABS

Yes No Don't Know 2. Wheels slowed by heavy braking (w/ABS?) If yes, Adjustment? _____

Yes No Don't Know 3. Wheels not in contact with ground (airborne) Over/Under report?

Yes No Don't Know 4. Yaw - Wheels scrubbing sideways If yes, Adjustment? _____

Yes No Don't Know 5. Ice/Snow/Hyrdoplaning/Burnout (wheels spinning relative to pavement)

Yes No Don't Know 6. In reverse or otherwise going backwards (spin?)

Yes No Don't Know 7. Data Limitations warn of special condition (i.e. GM last point recorded after AE?)

Vehicle Equipment Modifications

Yes No Don't Know 1. Factory Tire Size? If no, Adjustment? _____

Yes No Don't Know 2. Factory Axle Ratio? If no, Adjustment? _____

Yes No Don't Know 3. Control module reprogrammed for tire or axle size change?

Yes No Don't Know 4. Reprogrammed ACM? (Are Key cycles consistent, no signs of module tampering)

Delta V Data Special Circumstances/Adjustments

Yes No Don't Know 1. Missed DV before wakeup? (No for continuously running algorithms) (Small)
Yes No Don't Know 2. Accelerometer Zero shift? GM Adjusted Algorithm Delta V (AADV)?
Yes No Don't Know 3. Accelerometer peak value exceeded (clipping - accel trace flat-lined, DV>25-35?)
Yes No Don't Know 4. Offset collision – Need to adjust for Effective Mass Ratio?
Yes No Don't Know 5. Need to consider GROUND FORCES during crash? (For small ΔV's with braking)
Yes No Don't Know 6. Need to consider accelerometer RESOLUTION? (For small A's/ΔV's only).
Yes No Don't Know 7. Duration got **whole** crash? (Slope parallel to horizontal at end) If no, EDR DV is MINIMUM
Yes No Don't Know 8. Duration got **more** than whole crash (05+ Crown Vic)? If yes, Adjustment? _____
Yes No Don't Know 9. Need to consider **restitution** in this crash in closing speed formula?

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Have you considered <u>ALL</u> the possible uses of the data?

<u>SPEED DATA</u>

1. Fastest speed vehicle was traveling before the loss of control or braking? +/-4%
2. Speed at Impact? Start @ wheel speed = ground speed, use Worksheet to get range
3. Braking rate at different stages – transition from accel applied to normal braking to panic braking?
4. Departure speed (post impact)?
5. Can EDR data support or discredit witness statements?
6. Multiply Speed by time to calculate DISTANCE to impact at key points? Overlay speed data onto scene map. Check for witnesses at the location of illegal or negligent behavior.

<u>ACCEL PEDAL OR THROTTLE DATA</u>

1. Use ***<u>time</u> of accel pedal release*** to impact as first reaction to possible threat condition. Check time between accel pedal release and brake application – if fast, accel pedal release = reaction to perception of pending crash
2. Driver intent: ___0% coast ___1-19% slow down __20-30% maintain __31-50% accel ___ 51-100% heavy accel
3. Calculate **<u>distance</u>** to impact and overly on scene diagram. Use ***non-release or late release*** of pedal as ***possible*** evidence of obstructed vision, driver impairment by substance abuse, medical event, driver distraction (cell?)
4. Correlate to **witness statements** (heard engine race)

<u>BRAKING DATA</u>

1. Brake light on but no decrease in speed - hovering on pedal not applying pressure
2. Two brake switches – one on, other not on – hovering on pedal not applying pressure
3. Calculate distance to impact for first brake pedal touch using speed data, overlay on scene diagram or map, use late reaction or non-reaction as possible evidence of distracted driving. Overlay stop distance @ vehicle speed to avoid collision.

Memo: Avoid saying “NEVER hit brake” – suggest using “No evidence in EDR driver braked prior to impact”

<u>ABS ACTIVE DATA</u>

1. Indicates one wheel turning slower than others – can mean max braking intent OR loss of control. Brake does NOT have to be on for ABS to engage to try and relieve slowest wheel brake pressure.
2. No ABS on = driver did not push on pedal as hard as they could have.
3. Slow decel rate but ABS active may indicate ice, snow , hydroplaning – not enough friction available .

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STEERING ANGLE

1. Sign changes/overcorrection leading to rollover?
2. Swerve just before impact changing approach angle?
3. Swerve well before impact, in response to path intrusion, leading to going off road or crash into another object?
4. No visible input may indicate no clear alternative path available, or falling asleep, or medical event (check resolution before reaching conclusion of no input– early GM 16 degrees) .

YAW RATE

1. When/how far from impact did vehicle first begin to yaw?
2. Integrate yaw rate to calculate cumulative yaw angle/heading change prior to impact. Use to determine approach angle, also use to quantify speed under-reporting – divide speed by cosine of yaw angle = ground speed.

LATERAL ACCELERATION (PRE crash from stability control, not crash accelerometer)

Is vehicle in a yaw? G's show how far sideways it got – sideways if => drag factor 0.7G = 22ft/sec^2

LONGITUDINAL ACCELERATION (PRE crash from stability control, not crash accelerometer)

If in max braking, average of Longitudinal Accel should be equal to drag factor.

TIRE PRESSURE - Did low tire pressure contribute to heat buildup causing blowout followed by rollover?

TRACTION CONTROL

Did tires enough friction available to prevent wheel slip due to torque application?

ELECTRONIC STABILITY CONTROL ACTIVE -Use to indicate vehicle is yawing out of control

ROLL RATE OR ANGLE

Use to determine how quickly or slowly the trip or ramp roll initiating the rollover began. Integrate roll rate to get roll angle versus time. Be aware of max values, some clip at +/-240 degrees /sec or +/-60 degrees

ACM RECORDER ACCELERATION DATA

Use to determine if clipping occurred – look for flat lines at values at or above 32G (most are above 50)

RPM – Calculate Speed IF manual trans & clutch engaged OR auto trans torque converter is locked (not in 1st gear, not within 2 seconds of a shift, accel pedal applied 1-99%. Speed = RPM/trans gear/final drive gear/tire revs/mile*60min/hr = MPH

Example: 3200 rpm/0.7 overdrive/3.23 final drive/(750 revs/mile)*60 min/hr = 113 mph.

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Speed at Impact from Speed Data Worksheet

	MIN	MAX
Last Data Sample When wheel speed equals ground speed except for ABS slip	_____	_____
Braking between data sample and time of impact (if any) (Time Interval x Braking Rate) Braking rate: Drag Facor*G/1.466 fps/mph EX: 1.0 sec * 0.7g*32.2/1.466 = 15.4 mph If driver is still on accel pedal at last sample then allow 0.5 sec to move foot accel to brake pedal	-____	- 0
SLIP of wheel in ABS braking +5% (more during initial engagement)	+____	+____
Speedometer Error +/- 4% (unless there is data indicating it's better)	-_____	+_____
Range of Speeds at Impact (from the speed data alone)	_____ MIN	_____ MAX

Last sample before impact means last data point where wheel speed equaled ground speed except for stabilized ABS wheel slip, either during max ABS braking after initial engagement, or before wheel lockup in non-ABS vehicles. If you are not sure a vehicle is in max braking, run the worksheet both ways, first in max braking (usually results in most conservative value) and again without max braking.

Vehicles without ABS have little wheel slip (approx. 1%) until lockup is imminent, do not use 5% adjustment on non-ABS vehicles. When wheel lockup is imminent the speed drop from the prior point may appear to violate laws of physics, disregard these points and start from prior (last good) point.

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SPEED AT IMPACT FROM DELTA V WORKSHEET

Speed at Impact $= |V_3| Cos(\beta) - \Delta V_1$

Where

ΔV_1 = The Longitudinal ΔV of the Vehicle of Interest

$|V_3|$ = The absolute value of the post crash speed of the Vehicle of Interest (some use V_1')

β = (Beta) The angle measured between the approach and departure of the Vehicle of Interest. Beta can be between 0 and 360 degrees. For inline collisions cos β is 1..

This equation can be used in an inline or angular collision to calculate the Speed at Impact along the approach axis of either vehicle when the ΔV, the post collision speed and the departure angle Beta are known. It is necessary that the vehicle of interest is not side slipping (must be pointed in the +X direction) at impact. It is important to pay attention to the sign of the ΔV and the post crash velocity by carrying the sign of the cosine of Beta. For example if Beta = 135 degrees then Cos(135) = -.707

EXAMPLE: Vehicle travels 42 feet with an equivalent drag factor of 0.35
$|V_3| = \sqrt{30Df} = \sqrt{30*42*0.35} = 21.0$ mph

Departure angle β is 40 degrees relative to approach angle

ΔV_x = -31.0mph from EDR

$V_1 = 21.0mph*(Cos\ 40^o) – 31.0mph\ \Delta Vx$
$V_1 = 21.0*(.766) – (-31.0)$
$V_1 = 16.1 - - 31.0$
$V_1 = 47.1$ mph

Range of speeds at impact = +/-10% of ΔV plus any range due to drag factor uncertainty

Range = +/- 10% of -31 mph ΔV = +/- 3.1mph = 44.0 to 50.2 mph plus any range due to drag factor.

ΔV OF OTHER VEHICLE FROM SUBJECT VEHICLE ΔV

ΔV of other vehicle is inversely proportional to mass

$$\Delta V_1 = -\Delta V_2 \frac{W_2}{W_1}$$

Combine the two Delta V's to get closing speed:
Inline Closing Speed = (|ΔV1| + |ΔV2|)*(1/(1+e))
Combine closing speed with speed of known vehicle speed at impact to get speed of unknown.

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